# Properties of parallelograms

**1**

**a** $x = 115^\circ$ **b** $x = 101^\circ$ **c** $x = 30$ units **d** $x = 8$ units

**e** $x = 8$ units

**2**

**a** $x = 4, y = 69$

**b** $a = 57, b = 11$

**c** $x = 17, y = 7$

**d** $x = 4, y = 137, z = 43$

**e** $x = 7, y = 16$

**f** $x = 5, y = 8$

**g** $x = 4, y = 11, z = 8$

**h** $a = 89, b = 101$

**i** $k = 7, m = 8$

**j** $p = 5, q = 9$

**k** $w = 1, z = 3$

**l** $a = 12$

**3** $x = 2$ m

**4**

**a** $x = 9, y = 7$

**b** 20 units

**5**

**a**

**i** Slope of $\overline{AB} = \frac{1}{5}$, Slope of $\overline{CD} = \frac{1}{5}$

**ii** Slope of $\overline{AD} = -\frac{5}{3}$, Slope of $\overline{BC} = -\frac{5}{3}$

**b**

**i** No **ii** Yes **iii** No **iv** No

**6**

**a** No **b** No **c** Yes **d** No

**7** If a quadrilateral is a parallelogram, then its consecutive interior angles are supplementary.

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**8**

**To prove:** $\triangle ABC \cong \triangle CDA$

| | Statements | Reasons |
|---|---|---|
| **1.** | $ABCD$ is a parallelogram | Given |
| **2.** | $\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$ | Definition of a parallelogram |
| **3.** | $\angle BAC \cong \angle DCA$ and $\angle BCA \cong \angle DAC$ | Alternate Interior Angles Theorem |
| **4.** | $\overline{AC} \cong \overline{AC}$ | Reflexive Property of Congruence |
| **5.** | $\triangle ABC \cong \triangle CDA$ | Angle-Side-Angle Congruence |

**9**

**To prove: Opposite sides of a parallelogram are congruent**

| | Statements | Reasons |
|---|---|---|
| **1.** | $ABCD$ is a parallelogram | Given |
| **2.** | $\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$ | Definition of a parallelogram |
| **3.** | $\angle BAC \cong \angle DCA$ and $\angle BCA \cong \angle DAC$ | Alternate Interior Angles Theorem |
| **4.** | $\overline{AC} \cong \overline{AC}$ | Reflexive property of congruence |
| **5.** | $\triangle ABC \cong \triangle CDA$ | Angle-Side-Angle Congruence |
| **6.** | $\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$ | Corresponding parts of congruent triangles are congruent (CPCTC) |

**10**

**To prove:** Consecutive angles are supplementary.

| | Statements | Reasons |
|---|---|---|
| **1.** | $PQRS$ is a parallelogram | Given |
| **2.** | $\overline{PQ} \parallel \overline{SR}$ and $\overline{PS} \parallel \overline{QR}$ | Definition of a parallelogram |
| **3.** | $\angle P$ and $\angle S$, $\angle S$ and $\angle R$, and $\angle R$ and $\angle Q$ are supplementary. | Consecutive interior angles postulate |
| **4.** | $\angle P \cong \angle R$ and $\angle Q \cong \angle S$ | Congruent supplements theorem |

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**11**

**To prove: Opposite sides of a parallelogram are congruent**

| | Statements | Reasons |
|---|---|---|
| **1.** | $ABCD$ is a parallelogram | Given |
| **2.** | $\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$ | Definition of a parallelogram |
| **3.** | $\angle ABD \cong \angle CDB$ and $\angle ADB \cong \angle CBD$ | Alternate Interior Angles Theorem |
| **4.** | $\overline{BD} \cong \overline{BD}$ | Reflexive property of congruence |
| **5.** | $\triangle ABD \cong \triangle CDB$ | Angle-Side-Angle congruence theorem |
| **6.** | $\overline{AB} \cong \overline{CD}$ | Corresponding parts of congruent triangles are congruent (CPCTC) |
| **7.** | $\angle BAC \cong \angle DCA$ | Alternate Interior Angles Theorem |
| **8.** | $\triangle ABP \cong \triangle CDP$ | Angle-Side-Angle congruence theorem |
| **9.** | $\overline{BP} \cong \overline{DP}$ and $\overline{AP} \cong \overline{CP}$ | Corresponding parts of congruent triangles are congruent (CPCTC) |
| **10.** | $\overline{AC}$ bisects $\overline{BD}$ and $\overline{BD}$ bisects $\overline{AC}$ | Definition of a bisector |

**12**

**To prove: COnsecutive angles of a parallelogram are supplementary**

| | Statements | Reasons |
|---|---|---|
| **1.** | $PQRS$ is a parallelogram | Given |
| **2.** | $\overline{PQ} \parallel \overline{SR}$ and $\overline{PS} \parallel \overline{QR}$ | Definition of a parallelogram |
| **3.** | $\angle P$ and $\angle S$, $\angle S$ and $\angle R$, and $\angle R$ and $\angle Q$ are supplementary. | Consecutive interior angles postulate |

**13** The reason in line 6 uses the result of the proof, so it cannot be used.

**14** Line 3 should follow after line 1.

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## Additional questions

**15**

**a** True **b** False **c** False **d** True

**e** True **f** False

**16**

**a** Yes **b** No **c** No **d** Yes

**17**

**a** $x = 5, y = 8$ **b** $P = 44$ units

**18**

**a** $KF = 16$ **b** $m\angle H = 55^\circ$ **c** $m\angle H = 53^\circ$

**19** $y = 65$

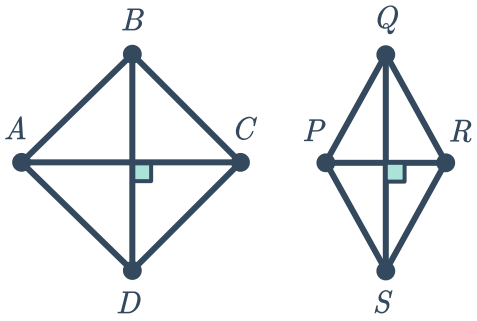
**20** $LN = 32$ units

**21**

**a** $\angle SPR \cong \angle PRQ$ **b** $\overline{PQ} \cong \overline{SR}$

**22**

**a**




**b** There is not enough information to conclude that $ABCD \cong PQRS$. We would need more information about the lengths of the sides or the diagonals in order to make that conclusion.

**c** Any of the following pieces of information about each quadrilateral: the diagonals bisect on another, they contain two pairs of opposite sides, they each contain opposite angles that are congruent, or they each contain adjacent angles that are supplementary.

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**23** Example proof

**To prove: $ABCD$ is a parallelogram**

| | Statements | Reasons |
|---|---|---|
| **1.** | $\overline{AB} \cong \overline{CD}$ | Given |
| **2.** | $\angle 1 \cong \angle 2$ | Given |
| **3.** | $\overline{AB} \| \overline{CD}$ | Converse of the alternate interior angles theorem |
| **4.** | $ABCD$ is a parallelogram | Definition of parallelogram |

**24** Option 1: It is possible that the garden could be a parallelogram, but not guaranteed. We would need one more piece of information such as whether or not the diagonals bisect one another in order to conclude the garden design is a parallelogram.
Option 2: Does not form a parallelogram, since the opposite angles are not supplementary.
Option 3: It is possible that the garden could be a parallelogram, but not guaranteed. We would need one more piece of information such as whether or not the opposite sides are parallel in order to conclude the garden design is a parallelogram.

**25** Example proof

**To prove: $E$ is the midpoint of $\overline{AC}$**

| | Statements | Reasons |
|---|---|---|
| **1.** | Parallelogram $ABCD$ | Given |
| **2.** | $\overline{AB} \| \overline{DC}$ | Definition of parallelogram |
| **3.** | $\overline{AB} \cong \overline{DC}$ | Definition of parallelogram |
| **4.** | $\angle 2 \cong \angle 4$ | Alternate interior angles theorem |
| **5.** | $\angle 1 \cong \angle 3$ | Alternate interior angles theorem |
| **6.** | $\triangle AEB \cong \angle CED$ | ASA congruence theorem |
| **7.** | $\overline{AE} \cong \overline{EC}$ | CPCTC |
| **8.** | $E$ is the midpoint of $\overline{AC}$ | Definition of midpoint |

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